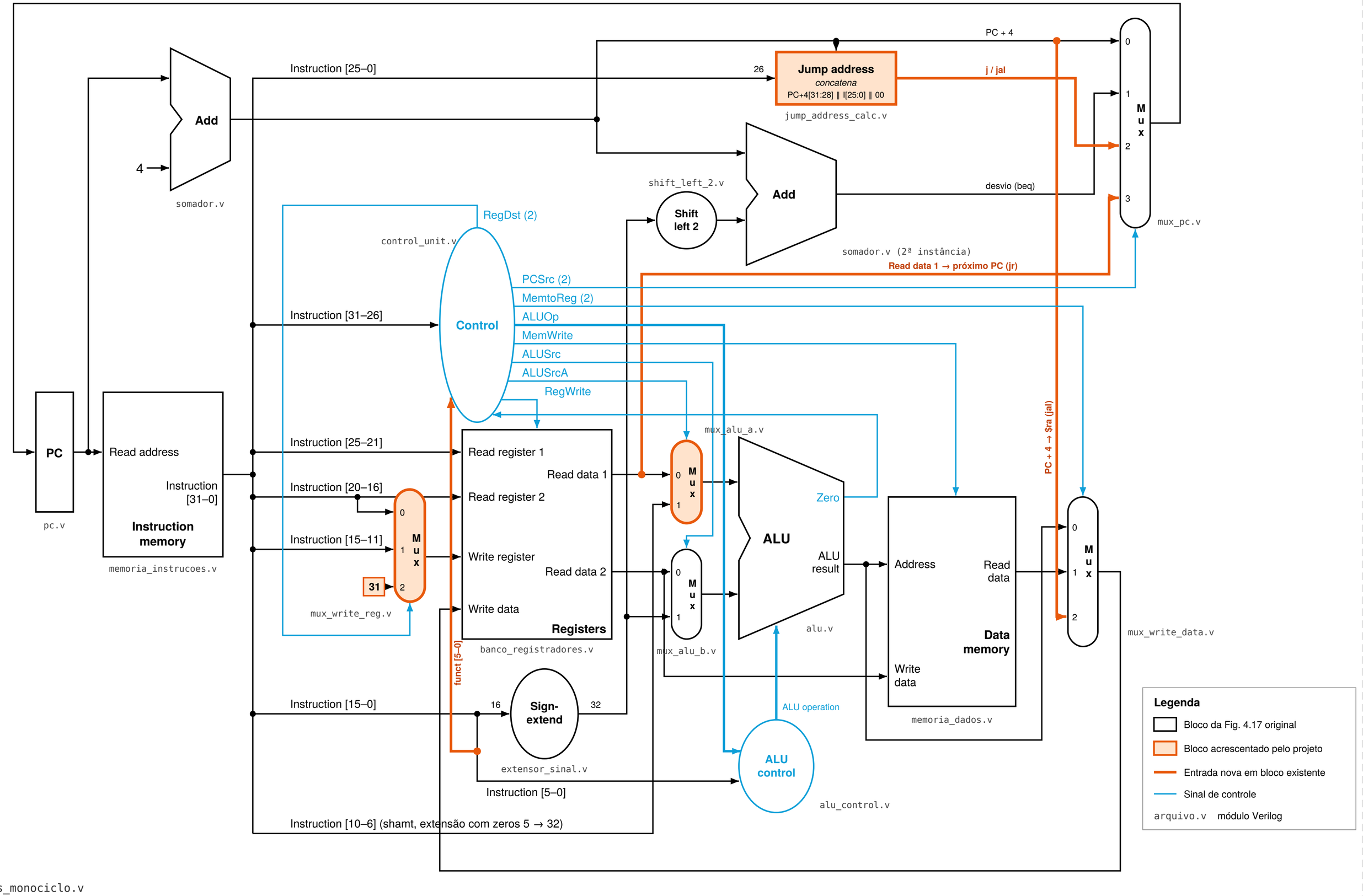
Add
4
somador.v
Instruction [25–0]
26
Jump address
concatena
PC+4[31:28] ‖ I[25:0] ‖ 00
jump_address_calc.v
PC + 4
j / jal
0
1
2
3
M
u
x
mux_pc.v
shift_left_2.v
Shift
left 2
Add
desvio (beq)
somador.v (2ª instância)
Read data 1 → próximo PC (jr)
RegDst (2)
control_unit.v
Instruction [31–26]
Control
PCSrc (2)
MemtoReg (2)
ALUOp
MemWrite
ALUSrc
ALUSrcA
RegWrite
PC
pc.v
Read address
Instruction
[31–0]
Instruction memory
memoria_instrucoes.v
Instruction [25–21]
Instruction [20–16]
Instruction [15–11]
31
mux_write_reg.v
Read register 1
Read register 2
Write register
Write data
Read data 1
Read data 2
Registers
banco_registradores.v
mux_alu_a.v
mux_alu_b.v
Zero
ALU
ALU
result
alu.v
Address
Read
data
Data
memory
Write
data
memoria_dados.v
PC + 4 → $ra (jal)
mux_write_data.v
funct [5–0]
Instruction [15–0]
16
Sign-
extend
32
extensor_sinal.v
Instruction [5–0]
ALU operation
ALU
control
alu_control.v
Instruction [10–6] (shamt, extensão com zeros 5 → 32)
Legenda
Bloco da Fig. 4.17 original
Bloco acrescentado pelo projeto
Entrada nova em bloco existente
Sinal de controle
arquivo.v módulo Verilog
mips_monociclo.v